1 FIG-213 — From-scratch failure distribution attribution

Group: MD-FROM-SCRATCH-FAILURE-ATTRIBUTION **Evidence chain:** manifested **Status:** descriptor/geometry attribution diagnostic; no causal claim

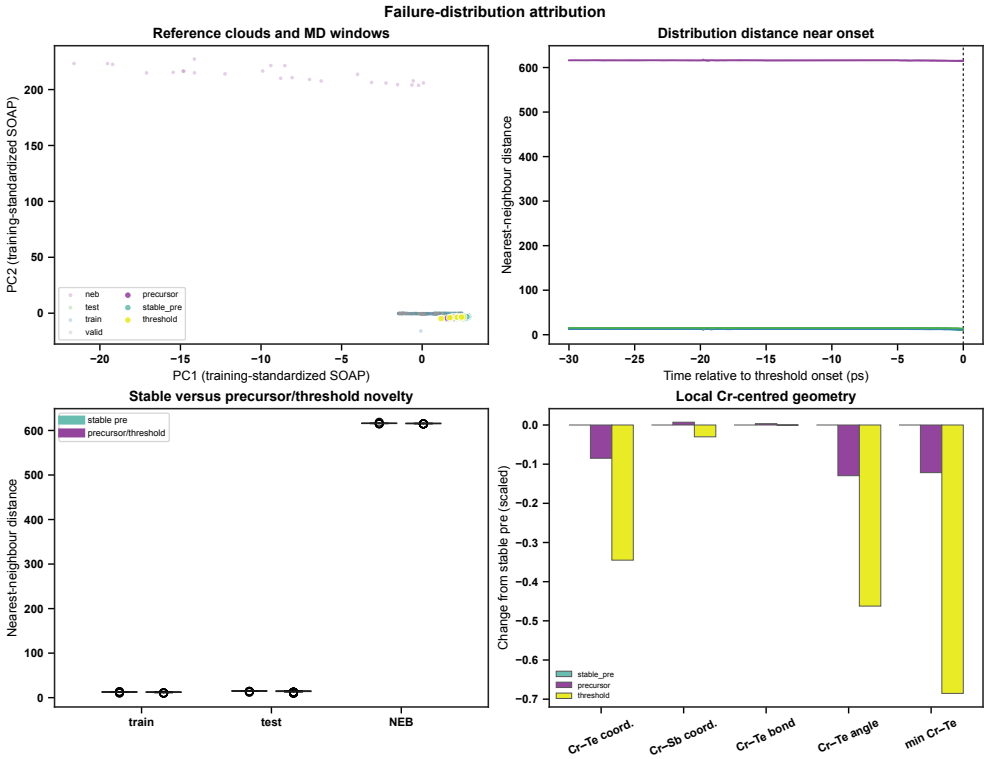


Figure 1: A, reference-embedding and MD-window distributions for the audited runs; B, embedding-distance distributions relative to the reference set; C, NEB-distance distributions for the available geometry windows; D, Cr-centered local geometry summarized by coordination and nearest-neighbor distance. The panels diagnose distributional novelty around the operational failure boundary; they do not identify a causal descriptor or calibrated uncertainty.

Results text draft. The audited windows show that unstable trajectories can occupy descriptor and local-geometry regions that differ from the reference distribution. The figure also preserves the overlap and heterogeneity across seeds and branches rather than reducing the outcome to a single novelty score.

Discussion and boundary. Descriptor novelty is used here as distributional evidence for follow-up triage only. It is not a calibrated model uncertainty, a failure classifier, or evidence that the descriptor shift caused the instability. The prepared DFT follow-up candidates were not submitted in this campaign.

Article placement. Proposed article location: Supplementary Information / from-scratch instability and failure-attribution audit; not yet a main-text causal claim.

Reviewer linkage: overall status PENDING.

- **REV-C9 (PENDING)**: Determine whether descriptor novelty is a useful diagnostic of the from-scratch failure boundary. *Completed:* Reference versus MD-window distributions and Cr-centered geometry are shown for the audited campaign. *Still required:* Establish calibration and independent validation before treating descriptor novelty as predictive or causal.

Evidence pointers.

- Data/processed: `/Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/results_all_replicates/frame_features.csv`
- Data/processed: `/Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/results_all_replicates/reference_embedding.csv`

- Plot/processing script: /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/plot_failure_attribution_all.py
- Plot Atlas and figure manifest: /Users/alina/Documents/Codex/plot_library; /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/figures/figure_manifest.json
- Asset SHA256: 59b81a790cf8552e6d50a52285e562de1970cc0f0d4bdf9c351e37a751a29659

Machine-collected evidence candidates (generated from the analysis manifest; do not edit this file).

- **Diagnostic report:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/results_all_replicates/RESULTS_DISCUSSION.md
- **Raw or processed data:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/results_all_replicates/diagnostic_manifest.json
- **Plot or processing script:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/plot_failure_attribution_all.py
- **Plot Atlas source:** /Users/alina/Documents/Codex/plot_library

Audit gap / next action: Run independent calibration and prospective validation before interpreting descriptor or geometry novelty as a predictor, uncertainty estimate, or causal mechanism.